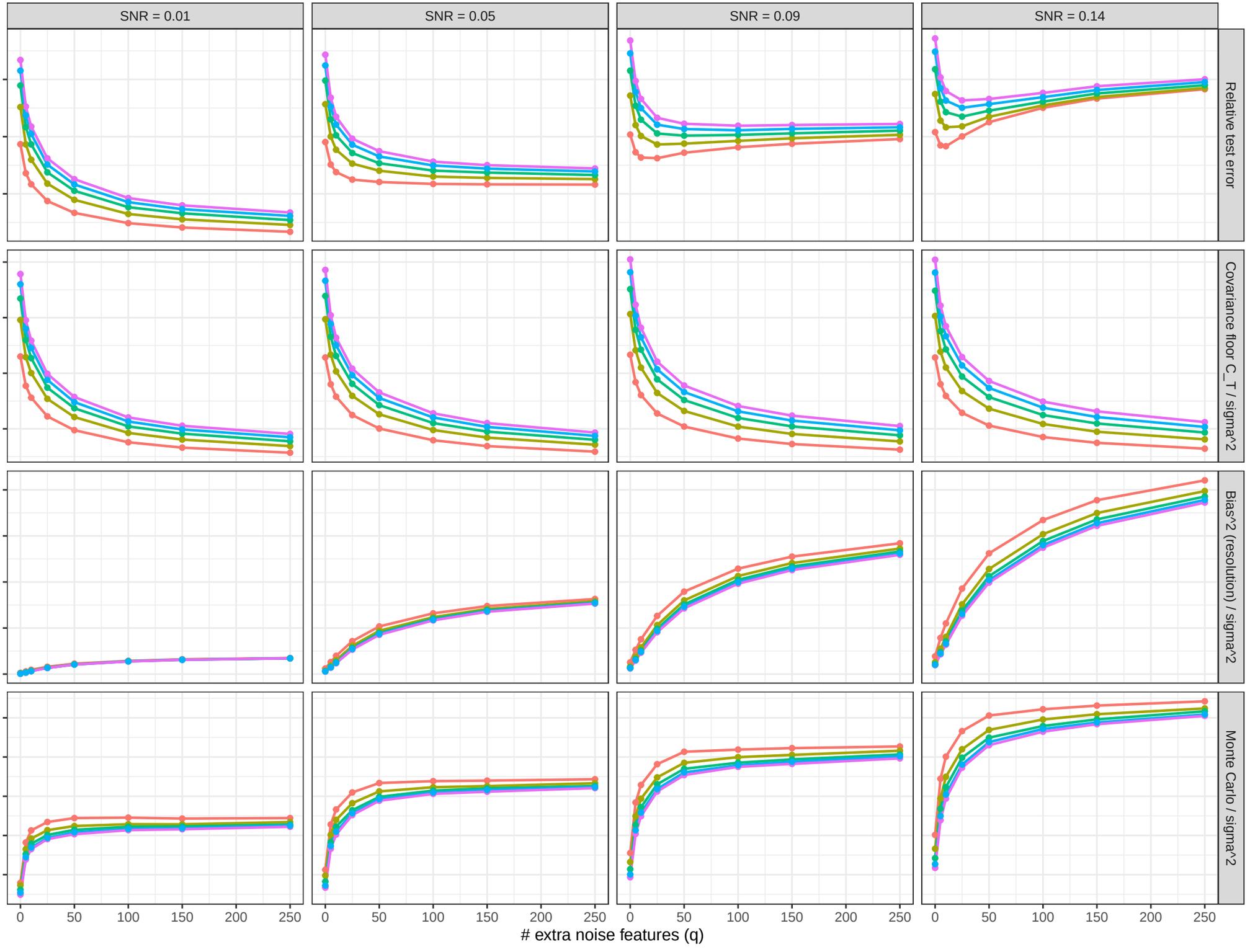
Covariance floor vs resolution as noise features are added

mtry is a fixed fraction of all p + q features.
n = 100, p = 5, r = 0.00, R = 150, B = 500, 3 X draws